

XINYI (CINDY) ZHOU

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EDUCATION

University of Southern California

M.S. in Computer Science; GPA: 4.0/4.0

January 2023 – December 2024

Los Angeles, CA

Southern University of Science and Technology

B.E. in Computer Science and Technology; **Outstanding Graduate Award**; GPA: 3.75/4.0

September 2018 – July 2022

Shenzhen, China

TECHNICAL SKILLS

Languages: TypeScript, Python, SQL, JavaScript, Java, C#, C++, HTML/CSS

Full-Stack Architecture: Next.js (SSR), React, Vue.js; FastAPI, REST APIs, XMPP Integration, Feature Flags (Unleash), Redis

Data/Infra: BigQuery, Snowflake, PostgreSQL, Amplitude, ETL; AWS (ECS, Lambda, S3, RDS/Aurora, CloudFront), Docker, CI/CD

ML/LLM: PyTorch, Hugging Face Transformers, DeepSpeed, Prompt Engineering, LLM Evaluation

Developer Operations & Tools: Better Stack (monitoring/alerting, logs), AI Agentic Workflows (Claude Code, Git Worktrees, Automated Error Detection/Debugging)

WORK EXPERIENCE

Software Engineer (Backend / AI Product) | Channel AI | Menlo Park, US

Feb 2025 – Present

- Led the implementation and rollout of a **personalization feature**, delivering **+45% revenue uplift** for mature cohorts within 4 days (Snowflake-validated) and an **approximately 4% increase** in daily messaging in the treatment cohort.
- Developed backend services for iOS, Android, and web clients deployed on **AWS**; implemented **XMPP** integration for real-time messaging, monitored production systems across **EC2**, load balancers, **RDS/Aurora**, **CloudFront**, and **Lambda**, and optimized **SQL queries** to improve latency, reliability, and scalability.
- Built an hourly **TypeScript** ETL pipeline on BigQuery (Amplitude exports) to refresh recommendation training data/features; added retries and windowed backfills to handle late-arriving events and maintain freshness.
- Built internal Trust & Safety tooling (text/image moderation + quality checks) integrating third-party classifiers; reduced manual review from 100% to **22%** by routing only **high-risk, policy-flagged** items to human review.
- Improved conversational and image-generation consistency across AI characters by iterating on prompts and building an **automated evaluation** loop (quantitative pass/fail tests + LLM-driven prompt refinement), while adding automated test coverage in existing **GitHub Actions** CI pipelines.
- Built internal **observability and debugging automation** for backend services, including a daily LLM-powered digest of **Sentry** and **Better Stack** errors and a **Slack bot** that launched **Claude** debugging sessions to investigate issues and open PRs with proposed fixes, reducing manual debugging effort and accelerating remediation.
- Designed and rolled out **database schema changes** and production migrations for new backend features, enabling product evolution while preserving data integrity, backward compatibility, and service reliability.

Software Engineer Intern | Veeva Systems, Vault Platform Reporting | Pleasanton, United States

May 2024 – August 2024

- Maintained a cloud-based reporting platform built with **Spring Boot** and **React** by resolving **11** defects, expanding automated test coverage with **JUnit** and **Jest**, and using **Jenkins** CI/CD pipelines to support reliable testing and deployment.
- Implemented a reporting data-freshness indicator using **React** and **JavaScript**, improving visibility into chart update recency for **1,000+ customers**.
- Resolved a critical **customer-reported** issue by fixing mapping-key collisions in a two-layer caching system caused by missing runtime conditions in cache key generation, restoring reliable report execution for **1,000+ customers**.

PUBLICATIONS

Xinyi Zhou, Zeinabsadat Saghi, Sadra Sabouri, Rahul Pandita, Mollie McGuire, Souti Chattopadhyay. “Cognitive Biases in LLM-Assisted Software Development.” *ICSE '26*, 2026. arXiv:2601.08045

Sadra Sabouri, Philipp Eibl, **Xinyi Zhou**, Morteza Ziyadi, Nenad Medvidovic, Lars Lindemann, Souti Chattopadhyay. “Trust Dynamics in AI-Assisted Development: Definitions, Factors, and Implications.” *ICSE '25*, 2025. doi:10.1109/ICSE55347.2025.00199

PROJECTS

Cognitive Biases in LLM-Assisted Software Development | Human-AI Interaction Researcher

January 2024 – October 2025

- Primary researcher for an ICSE 2026 paper on cognitive biases in LLM-assisted software development; led literature review, study design, recruitment, execution, and synthesis (arXiv:2601.08045).
- Ran mixed-methods research with developers (observations: n=14; surveys: n=22), combining behavioral coding and statistical analysis to measure bias in human-LLM workflows.
- Built a validated taxonomy of 15 bias categories (systematic analysis of 90 cognitive biases) and translated findings into concrete product/engineering guidelines for designing and evaluating LLM-driven systems with bias-mitigation guardrails.
- Measured bias prevalence in human-LLM workflows (48.8% biased actions; 56.4% tied to developer-LLM interactions), using results to define guardrail requirements and evaluation metrics.